PhD Math Camp

Formal Logic and Set Theory

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Outline

Introduction

The Role of Mathematics in Social Science Research

Why Mathematics is Essential:

- **Precision in Communication:**
  - Mathematics provides a precise language to describe and analyze complex social phenomena.
  - It allows us to define and quantify concepts such as probability, uncertainty, preferences, which are crucial in social science.
  - It allows us to be precise in what we mean when we develop and test theoretical claims.

Why we need math

- **Statistical Models:**
  - Mathematics underpins statistical models that are used to test hypotheses, measure relationships, and make predictions in social research.
  - Example: Regression models quantify the effect of one variable on another, helping to test theories.
- **Decision Making and Strategic Interaction:**
  - Mathematical models, such as game theory, provide frameworks for analyzing decision-making processes and strategic interactions among individuals and groups.
  - Example: The Prisoner's Dilemma models how individuals might not cooperate even when it's in their best interest, highlighting the tension between individual and collective rationality.

What Is a Proof?

Mathematics gives us two tools. Arithmetic and algebra let us **compute**; proof lets us **generalize**.

Proof

A finite sequence of statements, each of which is an assumption, a previously established result, or follows from earlier statements by a rule of inference — ending in the claim to be established.

A proof settles a property once and *in general*, so it can be used afterwards without being re-derived. Every theorem you will invoke in the methods sequence — the law of large numbers, the central limit theorem — is a result someone proved once so that everyone after them could simply use it.

The Vocabulary of Proof

In mathematical writing these four words are not interchangeable.

Assumption

Stated up front with clarity and taken to be true.

Proposition

A statement thought to be true given the assumptions made.

Theorem

A proved proposition.

Lemma

Like a theorem, but of little interest in itself — a stepping stone.

Besides assumptions, we also rely on proved properties and theorems to prove a theorem.

Approaches to Proofs

- **General deductive proof:** assume p and show q.
- **Proof by construction:** give an example to prove something exists.
- **Proof by counterexample:** give a counterexample to disprove a universal statement.
- **Proof by contradiction:** to prove p, assume $\neg p$ is true, and then prove that $\neg p$ implies both q and $\neg q$. Can also be used in science.
- **Proof by induction** (it is still deductive, though).

Mathematical Symbols

x^n	Exponentiation ("to the nth power")
$\sqrt[n]{x}$	Radical or nth root
$!$	Factorial
∞	Infinity
$\sum_{i=k}^{l} x_i$	Sum of x_i from index $i = k$ to $i = l$
$\prod_{i=k}^{l} x_i$	Product of x_i from index $i = k$ to $i = l$
$\ldots$	Continued progression
$\frac{d}{dx}$	Total derivative with respect to x
$\frac{\partial}{\partial x}$	Partial derivative with respect to x
$\int dx$	Integral over x
π	Pi, approximately 3.14159
e	Euler's number, approximately 2.71828

Set Theory Symbols

$\cup$	Set union
$\cap$	Set intersection
$\times$	Cartesian product of sets
$\setminus$	Set difference
A^c	Complement of set A
$\emptyset$	Empty (or null) set
$\in$	Set membership
$\notin$	Not member of set
$\mid$ or :	Such that
$\subset$	Proper subset
$\subseteq$	Subset

Logic and Set Notation Symbols

$f()$ or $f(\cdot)$	Function
$\{\}$	Delimiter for discrete set
$()$	Delimiter for open interval
$[]$	Delimiter for closed set
$\forall$	For all (or for every or for each)
$\exists$	There exists
$\Rightarrow$	Implies
$\Leftrightarrow$	If and only if
$\neg C$ or $\sim C$	Negation (not C)

Propositional and Predicate Logic

Why Formal Logic? Empiricism and Rationalism

Social science operates on a constant interplay between two modes of reasoning:

- **Empiricism:** Knowledge derived from observation and experience. This is the world of data, experiments, and statistical evidence. The core question is: *What does the world show us?*
- **Rationalism:** Knowledge derived from reason and logical inference. This is the domain of theory, axioms, and formal proofs. It establishes conclusions that must be true *if* the premises are true. The core question is: *What follows from our assumptions?*

Why Formal Logic? From Ideas to Testable Claims

A scientific claim is more than just an observation; it's a statement with a defined **truth value** (it can be true or false). Formal logic provides the precise, unambiguous language needed to:

- **Construct Valid Theories:** Ensure your theoretical arguments are sound.
- **Formulate Testable Hypotheses:** Translate a theoretical statement like "democracies don't fight each other" into a precise logical form that can be falsified.
- **Interpret Evidence Correctly:** Understand exactly what your statistical results do—and do not—imply about your theory.

Without formal logic, we risk ambiguous theories and invalid inferences.

Propositional Logic

Proposition (mathematical statement)

A declarative sentence that is either true or false, but not both.

Propositional logic deals with propositions and their interrelationships through logical connectives.

Basic Notation

Variables: p, q, r, etc.

- **Negation ($\neg$):** True if p is false.
- **Conjunction ($\wedge$):** True if both p and q are true.
- **Disjunction ($\vee$):** True if at least one of p or q is true.
- **Implication ($\rightarrow$):** False only when p is true and q is false.
- **Biconditional ($\leftrightarrow$):** True if both p and q are either true or false.

Necessary Conditions

Necessary condition

A condition q is **necessary** for a condition p if p cannot be true unless q is also true. Symbolically, $p \Rightarrow q$.

Example (In number sets). "If a number is an integer, then it is a rational number." Here "being a rational number" is necessary for "being an integer." In proofs, necessary conditions help identify what must be true for a theorem to hold.

Necessary and Sufficient Conditions

Necessary and sufficient

A condition p is **necessary and sufficient** for a condition q if q is true if and only if p is true. Symbolically, $p \Leftrightarrow q$.

Example (In number sets). "A number is prime if and only if it has exactly two distinct positive divisors: 1 and itself." This condition is both necessary and sufficient for "being prime."

- Demonstrating these conditions ensures complete characterization of the theorem.
- This equivalence is foundational in defining mathematical structures and relationships.

Truth Tables

Truth tables systematically list all possible truth values.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Conjunction ($\wedge$)

Logical Equivalence

Logical equivalence

Two statements are logically equivalent if they always have the same truth value.

Example. $p \to q \equiv \neg p \vee q$

Truth Table: $p \leftrightarrow q$ is Equivalent to $(p \to q) \land (q \to p)$

p	q	$p \to q$	$q \to p$	$(p \to q) \land (q \to p)$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

- The truth table shows that $(p \to q) \land (q \to p)$ and $p \leftrightarrow q$ have identical truth values in all possible scenarios.
- Therefore, $p \leftrightarrow q$ is logically equivalent to $(p \to q) \land (q \to p)$.

De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Contradictions and Tautologies

- **Contradiction.** A statement that is always false, e.g. $P \wedge \neg P$.
- **Tautology.** A statement that is always true, e.g. $P \vee \neg P$.

Propositional vs Predicate Logic

- **Propositional Logic:**
  - Basic unit: whole propositions (true/false statements).
  - Example: P: "It is raining."
  - Uses logical connectives: $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$.
- **Predicate Logic:**
  - Breaks propositions into objects and properties/relations.
  - Introduces **quantifiers**: $\forall$ (for all), $\exists$ (there exists).
  - Example: $\forall x\,(\mathrm{Cat}(x) \rightarrow \mathrm{Mammal}(x))$.

Key Difference

Propositional logic treats statements as indivisible; predicate logic looks inside them and expresses structure with quantifiers.

Quantifiers in Logic

- **Universal ($\forall$):** "for all" or "for every". Holds for all elements involved in a proposition.
- **Existential ($\exists$):** "there exists" or "for some". Exists at least one element.
- **Unique existential** ($\exists!$): "there exists exactly one".
- **Negated universal** ($\neg\forall$): equivalent to "there exists not" ($\exists\neg$).
- **Negated existential** ($\neg\exists$): equivalent to "for all not" ($\forall\neg$).

Note we often use $\nexists$ to express $\neg\exists$.

Predicates and Open Sentences

- **Predicate (propositional function).** A statement containing variables that becomes a proposition once the variables are specified. Example: $P(x)$: "x is even." Then $P(4)$ is true and $P(5)$ is false.
- **Domain of discourse.** The set of values over which a variable ranges.
- **Open sentence.** A predicate expressed as a statement with free variables; not true or false until variables are assigned or quantified.

Direct Proof Strategy Using Predicate Logic

Direct proof

Assume the premise (hypothesis) is true and use logical steps to show the conclusion follows. **Form:** for $p \to q$, assume p is true and demonstrate q is true.

Example (If n is an even integer, then n^2 is even). Let p be "$\exists k \in \mathbb{Z} : n = 2k$". The goal is to prove q: "$\exists m \in \mathbb{Z} : n^2 = 2m$".

> *Proof.* Assume $n = 2k$. Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$. Since $2k^2$ is an integer, n^2 is even. ∎

Proof by Contradiction Using Predicate Logic

Proof by contradiction

Assume the negation of the conclusion is true and show that this leads to a contradiction, thereby proving q is true.

Proof that $\sqrt{2}$ is irrational.

- Assume $\neg q$: "$\sqrt{2}$ is rational."
- This means $\sqrt{2} = \frac{a}{b}$, where a and b are coprime integers.[a]
- Squaring both sides gives $2 = \frac{a^2}{b^2}$, so $a^2 = 2b^2$.
- Thus a^2 is even, which implies a is even. Let $a = 2k$.
- Substituting into $a^2 = 2b^2$ gives $4k^2 = 2b^2$, so $b^2 = 2k^2$.
- Therefore b^2 is even, so b is even.
- This contradicts the assumption that a and b are coprime, so $\sqrt{2}$ must be irrational.

∎

[a] Two integers a and b are said to be coprime (or relatively prime) if the only positive integer that divides both of them is 1.

Valid Inference: The Core Idea

Inference rules are the scaffolding of logical arguments. They are patterns that guarantee a true conclusion if the premises are true. This is the foundation of moving from a theoretical hypothesis to an empirical conclusion.

Valid Inference Rules in Propositional Logic

- **Modus Ponens:**

$$\frac{X, \quad X \to Y}{\therefore Y}$$

If X is true and $X \to Y$ (if X then Y) is true, then Y must be true.

- **Modus Tollens:**

$$\frac{X \to Y, \quad \neg Y}{\therefore \neg X}$$

If $X \to Y$ is true and Y is false (i.e., $\neg Y$), then X must be false (i.e., $\neg X$).

- **Disjunctive Syllogism:**

$$\frac{X \vee Y, \quad \neg X}{\therefore Y}$$

If $X \vee Y$ (either X or Y) is true and X is false (i.e., $\neg X$), then Y must be true.

- **Hypothetical Syllogism (Transitivity):**

$$\frac{X \to Y, \quad Y \to Z}{\therefore X \to Z}$$

If $X \to Y$ and $Y \to Z$ are both true, then $X \to Z$ must be true.

Modus Ponens as a Logical Formula

Modus Ponens is a tautology

$$(X \wedge (X \to Y)) \to Y$$

X	Y	$X \to Y$	$X \wedge (X \to Y)$	$(X \wedge (X \to Y)) \to Y$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

The final column is true in every row.

Valid Inference: The Logic of Scientific Claims

Modus Ponens (The Way of Affirming)

$$\frac{P \to Q, \quad P}{\therefore Q}$$

If our theory is true ($P \to Q$), and we create the cause (P), we can deduce the effect (Q).

Modus Tollens (The Way of Denying)

$$\frac{P \to Q, \quad \neg Q}{\therefore \neg P}$$

If our theory is true ($P \to Q$), and we observe the effect is absent ($\neg Q$), we deduce the cause must have been absent ($\neg P$).

Hypothetical Syllogism (Chain of Inference)

$$\frac{P \to Q, \quad Q \to R}{\therefore P \to R}$$

Connects causal or logical chains. A cornerstone of theory building.

Disjunctive Syllogism (Process of Elimination)

$$\frac{P \vee Q, \quad \neg P}{\therefore Q}$$

If there are only two possibilities and we rule one out, the other must be true.

Inference Rules in a Social Psychology Experiment

Scenario: A researcher hypothesizes that exposure to money (*priming*) makes people more individualistic.

- Let P: A subject is primed with images of money.
- Let Q: The subject exhibits individualistic behavior (e.g., choosing to work alone).

Hypothesis (H_1)

$P \to Q$. If a subject is primed with money, they will act individualistic.

Example (Modus Ponens: The Logic of the Treatment Group). If H_1 is true ($P \to Q$), and we assign a subject to the treatment group (P), then we *predict* they will choose to work alone (Q). This is the core of experimental prediction.

Inference Rules in a Social Psychology Experiment (continued)

Example (Modus Tollens: The Logic of the Control Group). If H_1 is true ($P \to Q$), and a subject in the control group does *not* exhibit individualistic behavior ($\neg Q$), we can infer they were not primed with money ($\neg P$). This helps confirm the manipulation worked as intended. More powerfully, if a *treated* subject shows $\neg Q$, it is evidence *against* our hypothesis.

Example (Hypothetical Syllogism: Building a Causal Story). Suppose we also theorize that individualistic behavior (Q) leads to reduced requests for help from others (R).

- We have: $P \to Q$ (money prime $\to$ individualism)
- And: $Q \to R$ (individualism $\to$ less help-seeking)
- We can deduce a new, testable hypothesis: $P \to R$. Priming with money will cause subjects to seek less help.

Sets

Introduction to Sets

- **Set.** A collection of distinct objects, considered as a whole. Elements are listed within curly brackets, e.g. $A = \{1, 2, 3\}$.
- **Set-builder notation.** Describes a set by the properties its elements must satisfy.

Example. $A = \{x \in \mathbb{N} \mid x \text{ is even}\}$ denotes the set of all even natural numbers.

Equality of Sets and Subset

- **Equality of sets.** Two sets A and B are equal, denoted $A = B$, if they contain exactly the same elements — that is, every element of A is an element of B and every element of B is an element of A:

$$\forall x \, (x \in A \iff x \in B).$$

- **Subset.** A is a subset of B, denoted $A \subseteq B$, if every element of A is also an element of B:

$$A \subseteq B \iff \forall x \, (x \in A \to x \in B).$$

- If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

Important Sets

- **Empty set $\emptyset$.** The unique set containing no elements, $\emptyset = \{\}$. It is a subset of every set: $\emptyset \subseteq A$ for any A.
- **Universal set U.** Contains all elements under consideration in a particular context. All other sets are subsets of it.
- **Power set $\mathcal{P}(A)$.** The set of all subsets of A. If $A = \{1, 2\}$ then $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$, and $|\mathcal{P}(A)| = 2^{|A|}$.
- **Singleton.** A set containing exactly one element, e.g. $\{x\}$.

Operations on Sets

- **Union:** $A \cup B = \{x : x \in A \text{ or } x \in B\}$
- **Intersection:** $A \cap B = \{x : x \in A \text{ and } x \in B\}$
- **Difference:** $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$
- **Complement:** $A^c = \{x : x \notin A\}$
- **Cartesian Product:** $A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$

Properties of Sets

- **Algebraic properties.**
  - **Associativity:** $(A \cup B) \cup C = A \cup (B \cup C)$
  - **Commutativity:** $A \cup B = B \cup A$
  - **Distributivity:** $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- **De Morgan's Laws (sets).** $(A \cup B)^c = A^c \cap B^c$
  $(A \cap B)^c = A^c \cup B^c$

Proof of De Morgan's Law for Sets

Theorem

For any two sets A and B, $\quad (A \cup B)^c = A^c \cap B^c$.

Proof.

- Let $x \in (A \cup B)^c$. By definition of complement, $x \notin A \cup B$.
- This means x is not in either A or B. Therefore $x \notin A$ and $x \notin B$.
- Thus $x \in A^c$ and $x \in B^c$, which implies $x \in A^c \cap B^c$.
- Hence $(A \cup B)^c \subseteq A^c \cap B^c$.
- Now let $x \in A^c \cap B^c$. By definition $x \in A^c$ and $x \in B^c$, so $x \notin A$ and $x \notin B$.
- Therefore $x \notin A \cup B$, which implies $x \in (A \cup B)^c$.
- Hence $A^c \cap B^c \subseteq (A \cup B)^c$.
- Combining these, $(A \cup B)^c = A^c \cap B^c$.

$\blacksquare$

Applications

Number Sets

- **Natural Numbers ($\mathbb{N}$):** Positive integers from 1 upwards (sometimes includes 0).
- **Integers ($\mathbb{Z}$):** All positive and negative whole numbers, including zero.
- **Rational Numbers ($\mathbb{Q}$):** Numbers that can be expressed as a fraction $\frac{a}{b}$ with $a, b \in \mathbb{Z}$ and $b \neq 0$.
- **Real Numbers ($\mathbb{R}$):** All possible decimal values, including rational and irrational numbers.
- **Complex Numbers ($\mathbb{C}$):** Numbers of the form $a + bi$, where $i^2 = -1$.

Sets in Probability Theory

Probability theory uses the concept of sets to describe events and the sample space.

- **Sample space S.** The set of all possible outcomes in a probabilistic experiment.
- **Event.** A subset of the sample space S, representing a specific outcome or a group of outcomes.

Example (Coin toss). $S = \{\text{Heads}, \text{Tails}\}$. The event "getting Heads" is the set $E = \{\text{Heads}\}$.

Sample Space and Events

The sample space is a universal set

In the context of a probabilistic experiment, S plays exactly the role U plays in set theory, and an event E is any subset of it.

Example (Dice roll). For a dice roll, $S = \{1, 2, 3, 4, 5, 6\}$.

- $E_1 = \{2, 4, 6\}$ (rolling an even number)
- $E_2 = \{1, 2, 3\}$ (rolling a number less than 4)

Union and Intersection of Events

Union and intersection of events

The **union** $A \cup B$ is the event that occurs if either A, B, or both occur. The **intersection** $A \cap B$ is the event that occurs only if both A and B occur.

Example (Dice roll). If $A = \{1, 2, 3\}$ and $B = \{2, 4, 6\}$, then $A \cup B = \{1, 2, 3, 4, 6\}$ and $A \cap B = \{2\}$.

Mutually Exclusive Events in Probability

Mutually exclusive events

Two events are mutually exclusive if they cannot occur simultaneously, that is, if $A \cap B = \emptyset$.

Example (Rolling a die). Let A be "rolling an even number" ($A = \{2, 4, 6\}$) and B be "rolling an odd number" ($B = \{1, 3, 5\}$). Then

$$A \cap B = \emptyset,$$

so these events cannot occur simultaneously in a single die roll.

Cartesian Product and Strategy Profiles

Each prisoner has two possible strategies: **Cooperate (C)** or **Defect (D)**, so the strategy set for each prisoner is $S = \{C, D\}$.

The set of all possible strategy profiles is the Cartesian product:

$$S_1 \times S_2 = \{C, D\} \times \{C, D\} = \{(C, C), (C, D), (D, C), (D, D)\}$$

Each element represents a possible outcome of the game: (C, C) both cooperate; (D, C) prisoner 1 defects while prisoner 2 cooperates.